

Pushpak Vijay Katkhede

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SUMMARY

Software Engineer with 3+ years building backend services, distributed systems, and cloud-native applications at IBM and Oregon State University. Experienced designing REST APIs, automation frameworks, enterprise integrations, and containerized microservices on Docker and Kubernetes. Published author at ICSE 2026 (ACM/IEEE) on REST API defect analysis. Open to remote or relocation.

TECHNICAL SKILLS

Languages: Python, C++, Bash, Java

Frontend: HTML, JavaScript, React.js, CSS, TypeScript

Backend: Node.js, Express.js, Flask, FastAPI, REST APIs, SOAP, Postman, Unit Testing

Databases: SQL (MySQL, PostgreSQL, Oracle), NoSQL (MongoDB, AWS DynamoDB)

DevOps/Cloud: Docker, Kubernetes, Jenkins, Terraform, CI/CD, Git, Linux, AWS (EC2, Lambda, S3), Azure (Entra ID), Agile, Scrum

Data/AI: PyTorch, TensorFlow, NLP, Embeddings, LLMs, Transfer Learning, Topic Modeling

WORK EXPERIENCE

Oregon State University

Software Engineer

Corvallis, OR, US | May 2025 - Present

- Engineered 20+ Python and SQL integration pipelines synchronizing data attributes across enterprise systems (Workday, Saviynt, Entra ID, Grouper) and mapping 850+ access configuration groups across 125+ applications, de-risking a large-scale HR platform migration from on-prem to cloud.
- Designed a self-service portal catering to 100K+ users, handling account activation and identity self-service.
- Architected an internal tool and its reusable Python SDK for API orchestration, authentication, and logging, automating daily operations and cutting development effort by 50%.
- Delivered and maintained 10+ production system integrations using SOAP, REST, and SQL connectors, enabling automated data synchronization and cross-platform reconciliation across enterprise systems.
- Developed an automated testing framework validating data consistency and system state across 9 enterprise integrations, eliminating 15+ hours/week of manual verification.

Research Assistant

Corvallis, OR, US | Dec 2024 - May 2025

- Led a research initiative to develop a categorized taxonomy of bugs in RESTful APIs, establishing a benchmark for evaluating automated testing tools and improving bug localization in source code.
- Designed and implemented a scalable research pipeline for bug extraction using GitHub API mining over 200K issues, integrating custom tokenization, BERT embeddings, and StarCoder2 LLMs for representation learning and topic modeling.
- Built a full-stack framework to auto-grade student APIs, database, and UI, cutting grading errors by 90% and eval time by 50%.

IBM India

Software Engineer

Bangalore, KA, IN | Oct 2019 - Jul 2022

- Designed and built a self-service trial environment for observability pipelines with automated infrastructure provisioning using Terraform, enabling 50+ enterprise customers to run evaluations that converted into \$3M in signed contracts.
- Led a cross-functional team of 6 to design and build a suite of simulators for API interaction, microservice failures, and user behavior simulation enabling fault-injection for assessment of the complex observability pipeline.
- Migrated a large-scale legacy web application from vanilla JavaScript to React, implementing a component-based architecture with Redux and Context API, improving CSAT by 23%.
- Improved application performance by 30% by refactoring Node.js and Java services using event-driven architecture, lazy loading, asynchronous programming, and caching.
- Containerized and orchestrated 15+ microservices serving 500K+ users, integrating them into CI/CD pipelines with automated deployment and achieving 99% fault-tolerant deployments with monitoring dashboards.

PUBLICATIONS

- Defects4REST: A Benchmark of Real-World Defects to Enable Controlled Testing and Debugging Studies for REST APIs**, ACM/IEEE International Conference on Software Engineering (ICSE), 2026.
- Live Assistance and Tracking System (LATS) using Cloud and IoT**, International Journal of Computer Sciences & Engineering (IJCSE), 2019.

EDUCATION

Master of Science, Computer Science

Sep 2022 - Dec 2024

Oregon State University

Corvallis, OR, US

• **GPA:** 3.75/4.0

• **Coursework:** Algorithms Design and Analysis, Software Engineering, Parallel Programming, Deep Learning, Project Management

Bachelor of Engineering, Computer Engineering

Aug 2015 - Jun 2019

Savitribai Phule Pune University

Pune, MH, IN

• **GPA:** 8.10/10.0

• **Coursework:** Data Structures, Database Management Systems, Operating Systems, Machine Learning, Software Design and Modeling

ACADEMIC PROJECTS

Ask chAMP! - RAG Chatbot (Python, Claude API, Vector DB)

- Developed a Retrieval-Augmented Generation (RAG) chatbot with multi-source ingestion and persistent caching, using a vector database for semantic retrieval and the Claude API for response generation.

Multi-tier Research Manager powered by Microservices (Python, FastAPI, React, PostgreSQL, Docker, Kubernetes)

- Prototyped a collaborative, multi-tier platform envisioned as a one-stop workspace for research teams to manage day-to-day tasks, progress, and feedback, with role-based access control (RBAC) across containerized microservices.